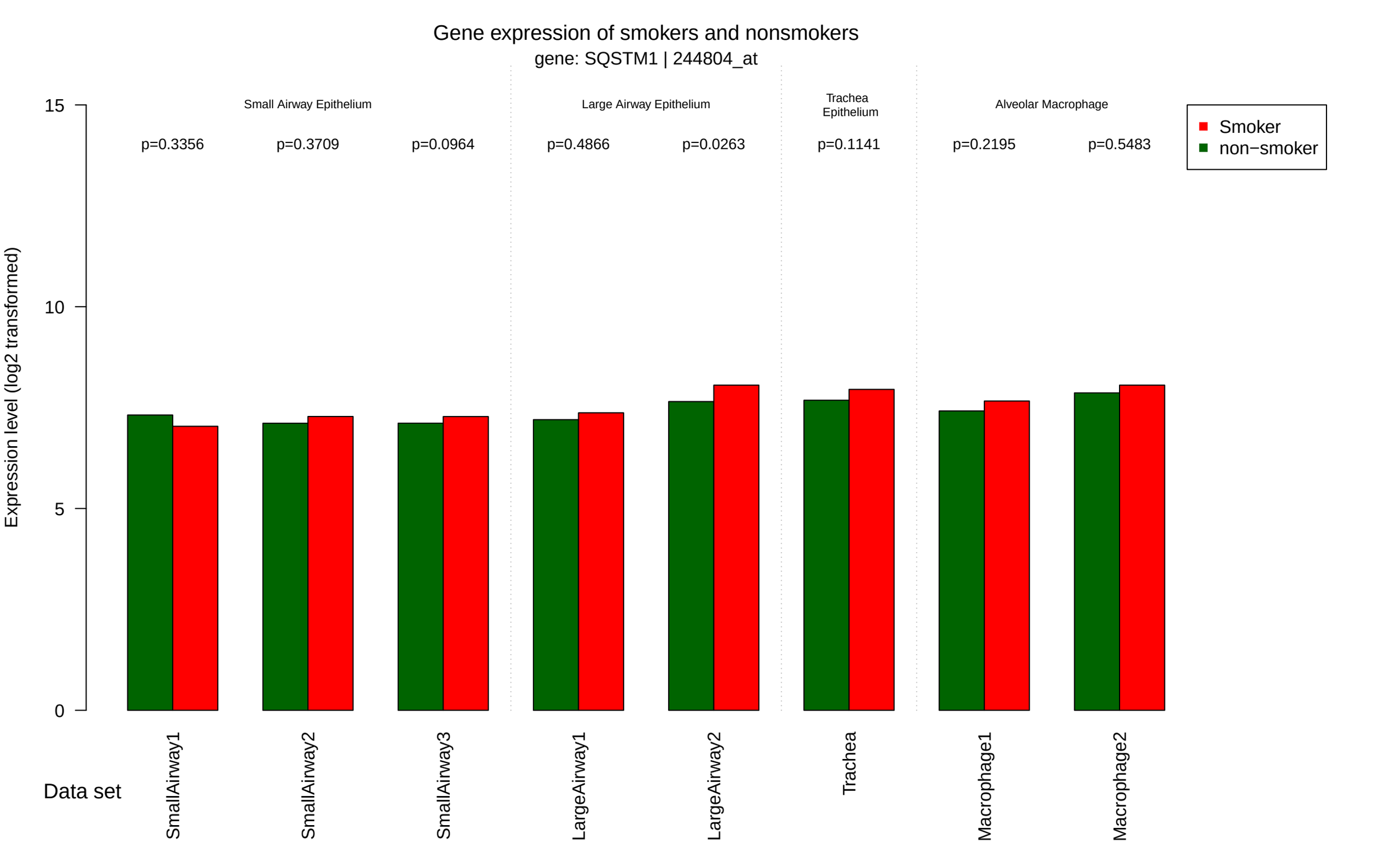
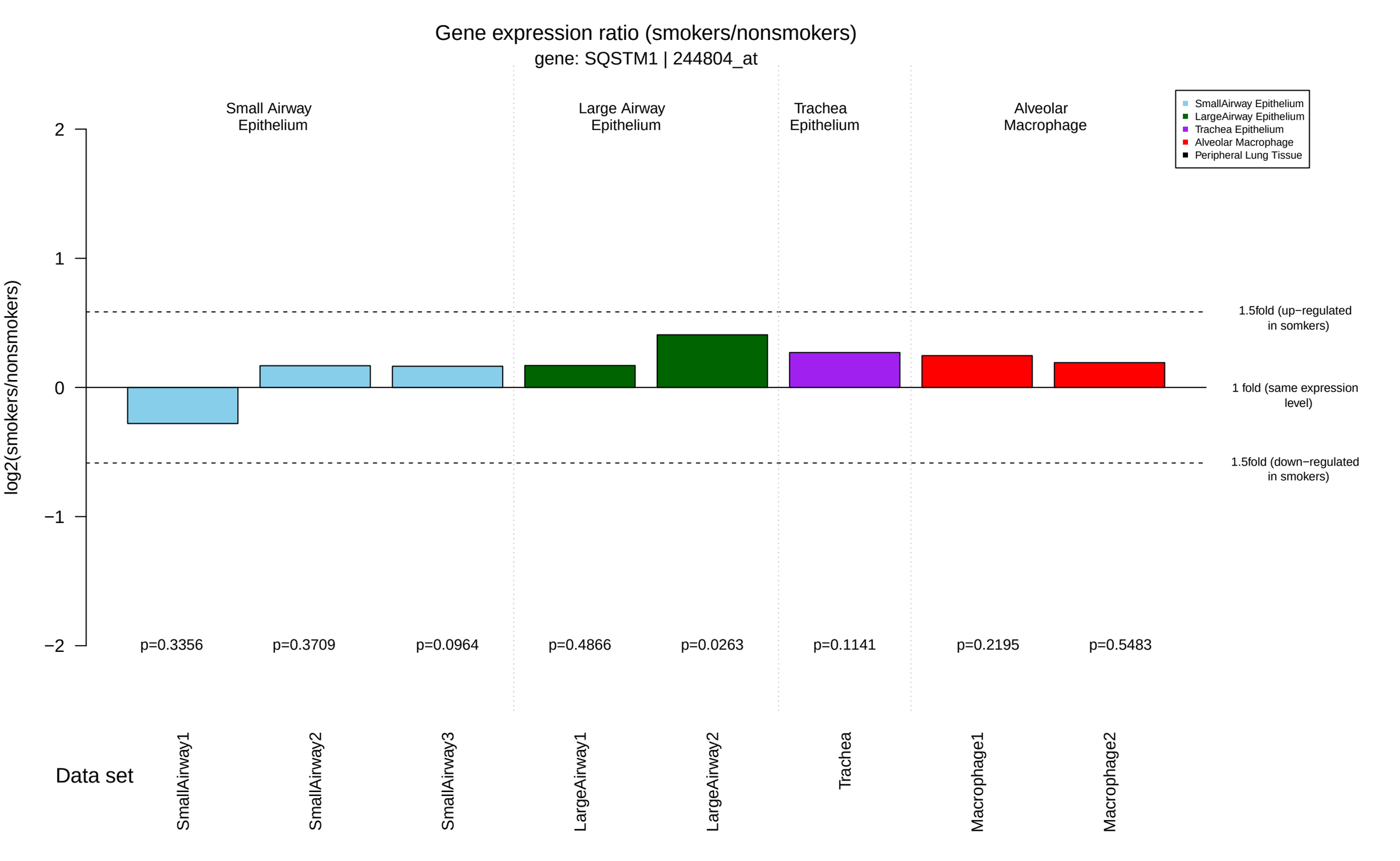
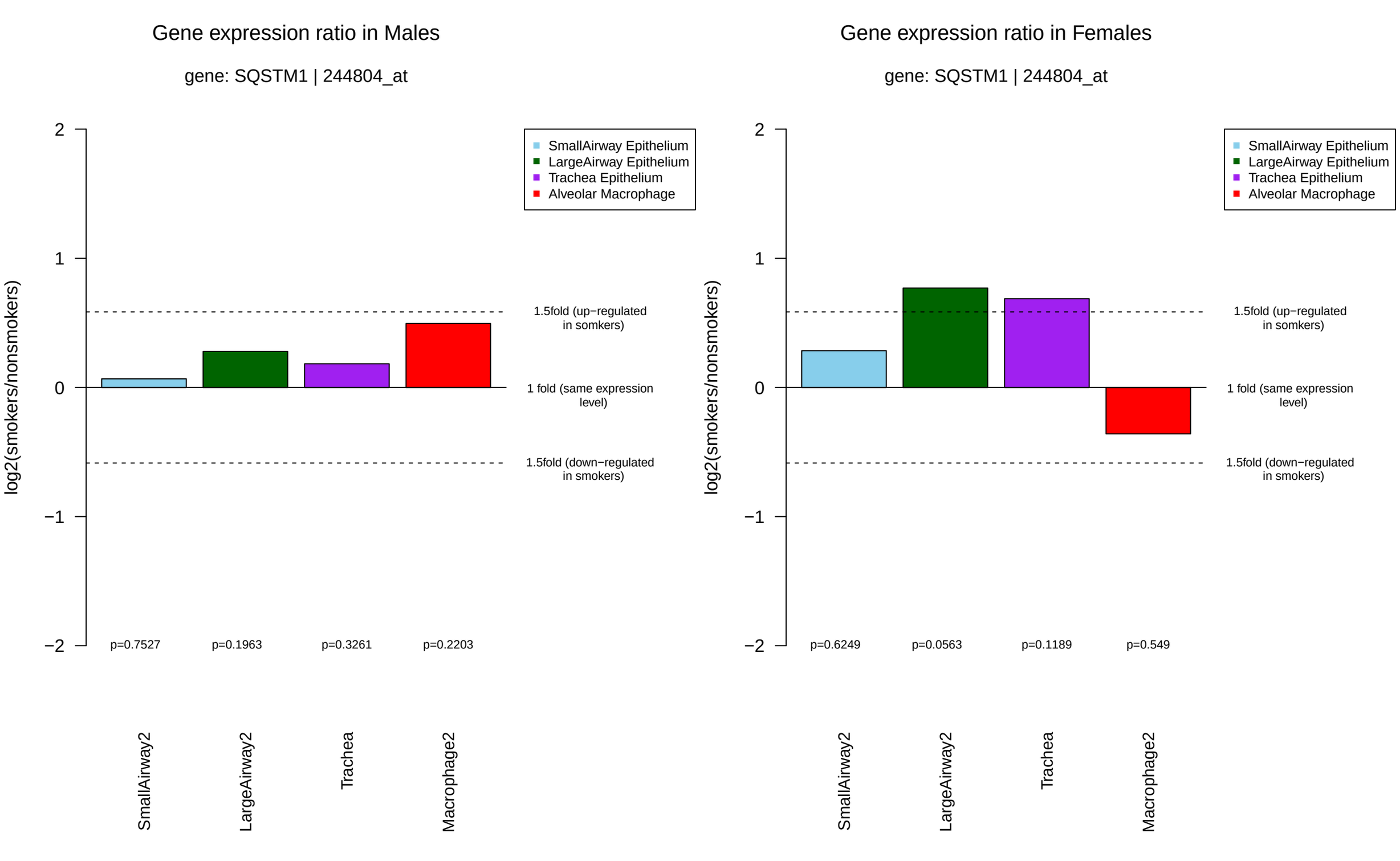
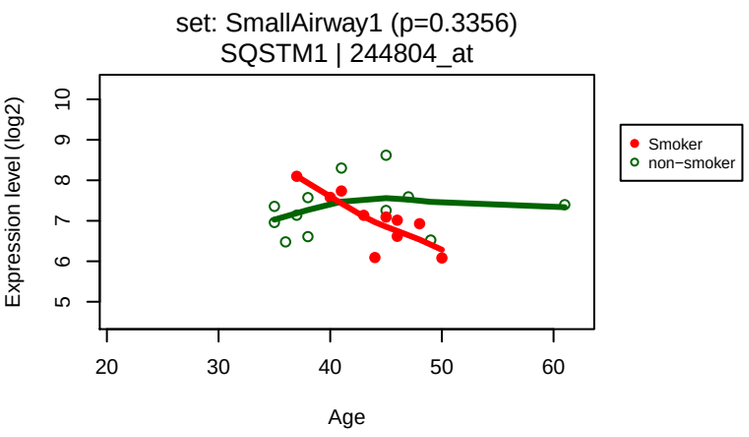
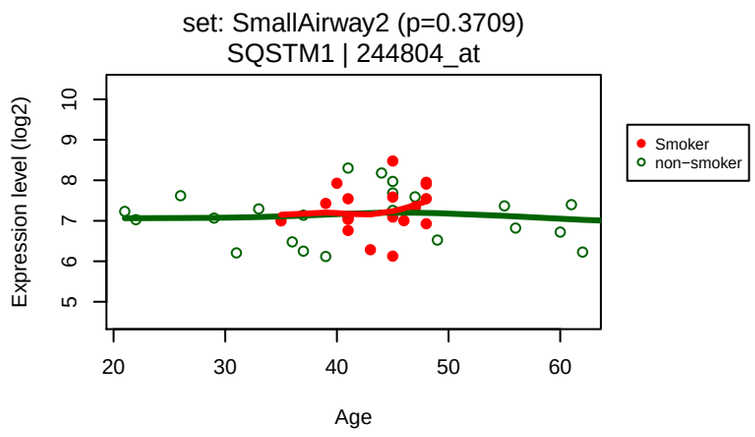
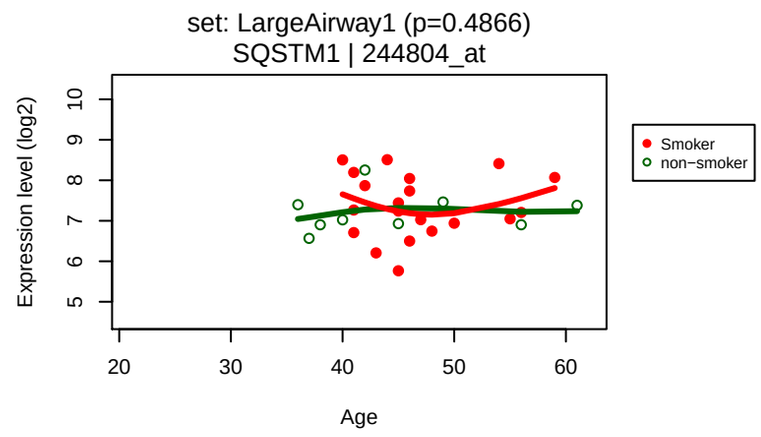

Gene expression of smokers and nonsmokers
gene: SQSTM1 | 244804_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3356
p=0.3709
p=0.0964
p=0.4866
p=0.0263
p=0.1141
p=0.2195
p=0.5483
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

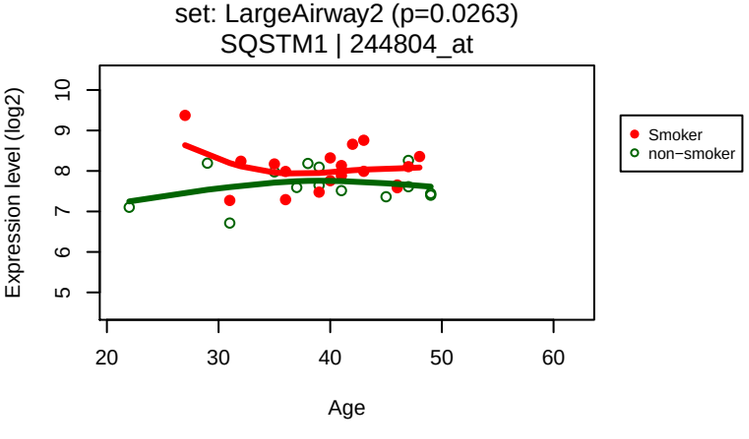

Gene expression ratio (smokers/nonsmokers)
gene: SQSTM1 | 244804_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3356
p=0.3709
p=0.0964
p=0.4866
p=0.0263
p=0.1141
p=0.2195
p=0.5483
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

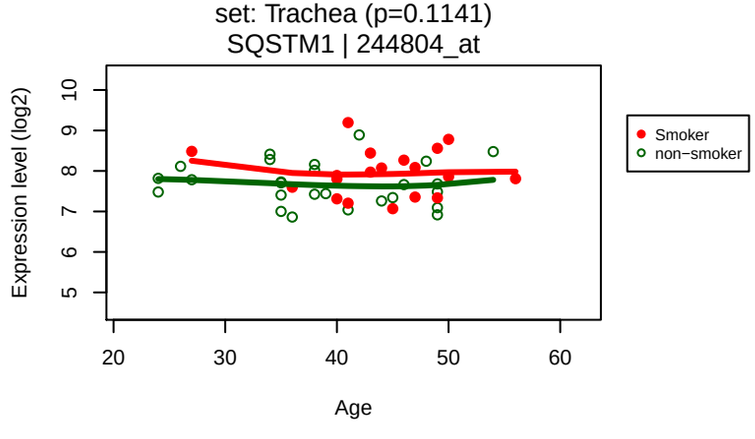

Gene expression ratio in Males
gene: SQSTM1 | 244804_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7527
p=0.1963
p=0.3261
p=0.2203
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SQSTM1 | 244804_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6249
p=0.0563
p=0.1189
p=0.549
SmallAirway2
LargeAirway2
Trachea
Macrophage2

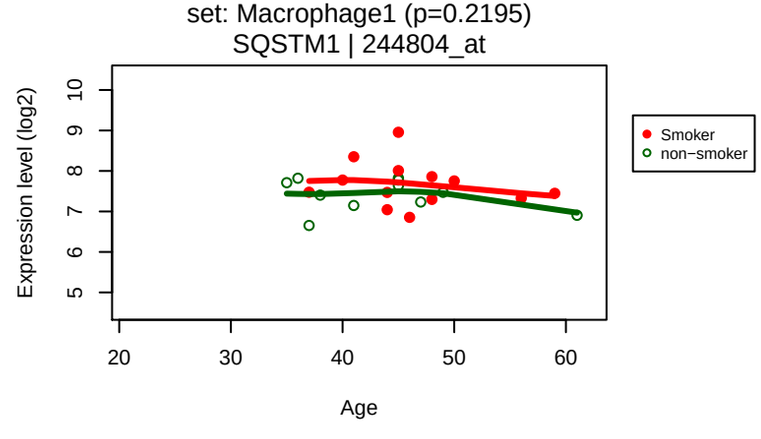

set: SmallAirway1 (p=0.3356)
SQSTM1 | 244804_at
Expression level (log2)
Age
Smoker
non−smoker

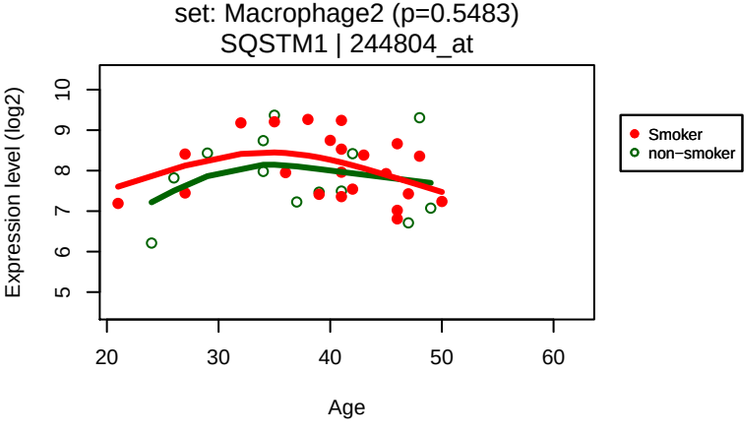

set: SmallAirway2 (p=0.3709)
SQSTM1 | 244804_at
Expression level (log2)
Age
Smoker
non−smoker

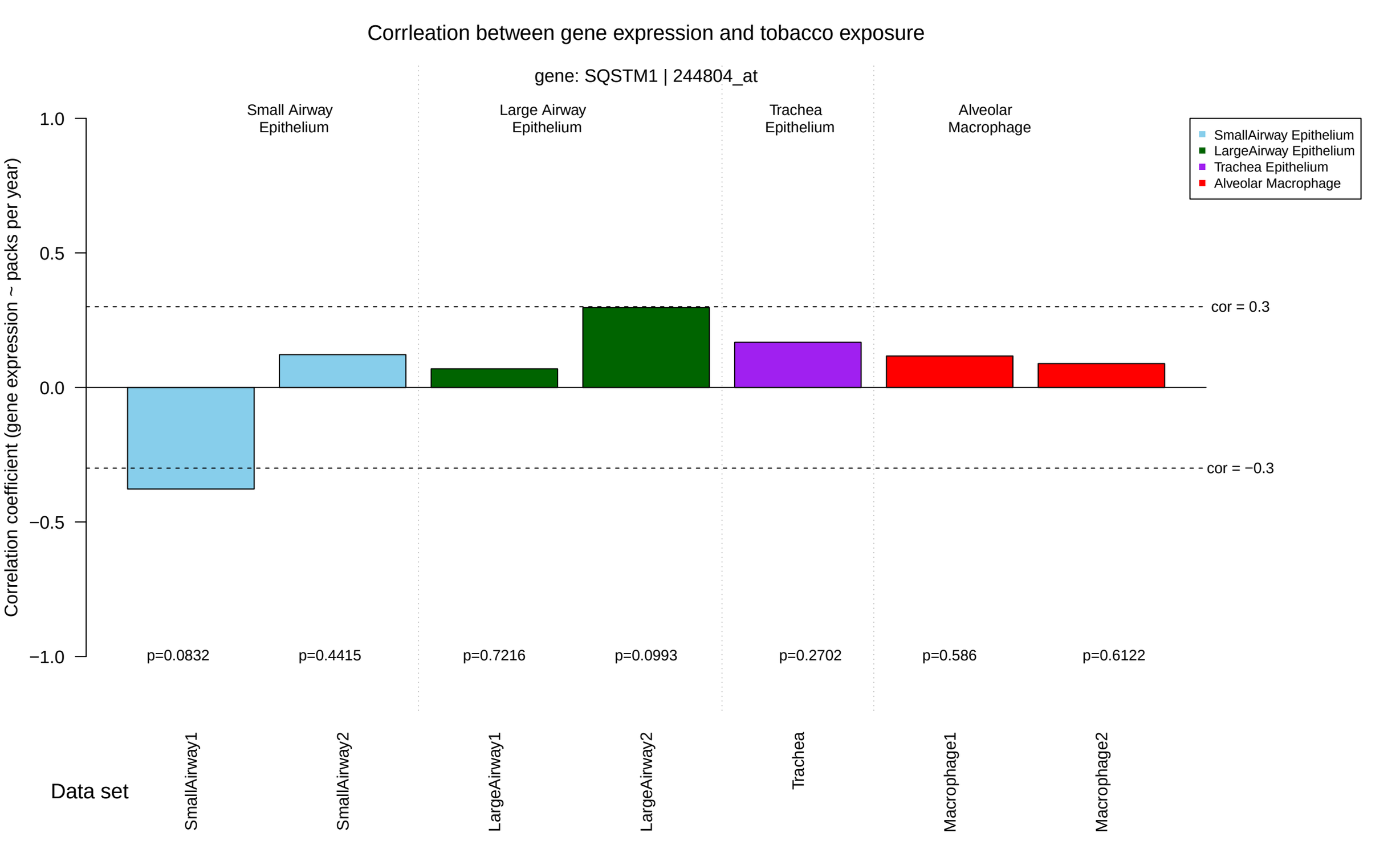

set: LargeAirway1 (p=0.4866)
SQSTM1 | 244804_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0263)
SQSTM1 | 244804_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1141)
SQSTM1 | 244804_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2195)
SQSTM1 | 244804_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5483)
SQSTM1 | 244804_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SQSTM1 | 244804_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0832
p=0.4415
p=0.7216
p=0.0993
p=0.2702
p=0.586
p=0.6122
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2